

ZHANLI LI (李展利)

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EDUCATION

Zhongnan University of Economics and Law

September 2023 – June 2027

Wenlan School of Business (Honors College), B.Sc. in Digital Economy

Weighted Avg: 93.73/100 Rank: 2/80 2025 National Scholarship

Research Interests: LLMs, Agentic Training, Deep Learning, Causal Inference

PUBLICATIONS

Zhanli Li, Huiwen Tian, Lvzhou Luo, Yixuan Cao, Ping Luo. *DeepRead: Document Structure-Aware Reasoning to Enhance Agentic Search*. *KDD 2027* (CCF-A), **Resubmit**. [First Author]

Zhanli Li, Yixuan Cao, Lvzhou Luo, Ping Luo. *Navigating Large-Scale Document Collections: MuDABench for Multi-Document Analytical QA*. *ACL 2026 Findings* (CCF-A). [First Author]

Zhanli Li, Zichao Yang. *ESG Rating Disagreement and Corporate Total Factor Productivity: Inference and Prediction*. *Finance Research Letters*, vol. 78, p. 107127, 2025. (CAS Q1 Top, JCR Q1, IF: 6.9). [First Author]

RESEARCH EXPERIENCE

NVIDIA Nemotron Model Reasoning Challenge *Silver Medal*

June 2026

On a new **abstract puzzle reasoning benchmark** built around NVIDIA Nemotron open reasoning models, we used **Nemotron-3-Nano-30B-A3B** as the backbone and explored ways to improve reasoning accuracy under the competition-required **LoRA fine-tuning** setting. The benchmark emphasized multi-step deduction under strict prefix constraints, so we built an **agentic reasoning** framework that repeatedly generated candidate solutions, checked them against the puzzle rules, and fed failed attempts back into the next search round. We further optimized the **rule orchestration** process by clustering and ordering discovered rules into a **low-entropy** representation: redundant heuristics were merged or pruned, rule triggers were made explicit, and solver priorities were arranged to expose stable, learnable decision paths. This turned ad hoc discoveries into a compact rule system of reusable **solvers**, making the rules easier for the model to learn, retrieve, and reuse while preserving prefix-reasoning consistency and answer verifiability across related puzzles, ultimately achieving a score of **0.86**.

▷ **Kaggle Leaderboard Silver Medal**, ranked **119/4182** globally on the final leaderboard.

Agentic Training for Real-World Data Analysis[‡]

March 2026 – Present

Role: Project Leader

Existing evaluation benchmarks for data analysis agents typically rely on **fixed reference answers**, making it difficult to capture **non-canonical yet valuable insights** generated in open-ended analytical settings, thereby systematically underestimating their true capability. To address this, we construct a **large-scale interactive data lake environment** that supports autonomous exploration under realistic data distributions. Built on this environment, we train **Qwen3.5-9B** as a **verifier agent** on **24 A100 GPUs** to perform **fine-grained verification** of each insight produced by the data agent, assessing both correctness and analytical value. We further integrate the trained verifier into an **iterative agent harness**, forming a “**generate-verify-feedback**” loop that continuously yields **high-value insights** in real-world scenarios.

▷ Selected for the **Zhongguancun Academy Shenlan Program**; intended submission to *ICLR 2027*.

DeepRead: Document Structure-Aware Reasoning to Enhance Agentic Search[†] *October 2025 – February 2026*

Role: Project Leader

Existing agentic search frameworks often treat long documents as flat chunk collections, ignoring native **hierarchical structures** and **sequential logic**. We propose **DeepRead**, a structure-aware document reasoning agent that preserves layout fidelity via OCR, constructs **paragraph-level coordinates**, and equips the

LLM with two synergistic tools: **Retrieve** and **ReadSection**. This induces an emergent “locate-then-read” reasoning paradigm, significantly mitigating context fragmentation issues inherent in conventional retrieval. Across four benchmarks covering diverse document types, DeepRead outperforms **Search-o1** style agentic search baselines by an average of **10.3%**.

▷ **Resubmit to KDD 2027 (CCF-A)**. The preprint was reported by the prominent tech media **New Wisdom (新智元)** and received over 130 saves on Xiaohongshu.

MuDABench: Multi-Document Analytical QA Benchmark and Multi-Agent Framework[†] *April 2025 – December 2025*

Role: Project Leader

Large-scale document collections contain rich knowledge, yet scalable multi-document analytical QA across **thousands** of long documents remains challenging. We constructed one of the **largest** multi-document analytical QA benchmarks to date (**80k+** pages, **332** analytical questions, **4,964** structured intermediate facts) using **distant supervision** and expert verification, enabling systematic evaluation of multi-document retrieval, cross-document aggregation, and numerical reasoning. Experiments reveal that standard retrieval services struggle in this setting: OpenAI’s **File Search** achieves a very low best accuracy. Our proposed **Multi-Agent** workflow raises this substantially. Further error analysis indicates that the primary bottleneck lies in the stability and high-precision localization of **single-document information extraction**.

▷ **Published in ACL 2026 Findings (CCF-A)**.

[†]Conducted under the supervision of Associate Professors Yixuan Cao and Ping Luo, Key Laboratory of Intelligent Information Processing, Institute of Computing Technology, Chinese Academy of Sciences

[‡]Conducted under the supervision of Assistant Professor Wentao Zhang, Peking University & Zhongguancun Academy

INTERNSHIP EXPERIENCE

Beijing Paoding Technology Co., Ltd. — Research Dept.

June 2025 – February 2026

Role: AI Research Intern **Mentors:** Ping Luo, Yixuan Cao

Project: Improving Context Retrieval in Production-Grade RAG Systems

Leveraged the company’s proprietary **document parsing** model PDFlux and integrated its outputs into the commercial RAG product ChatDOC, focusing on mitigating **context-missing** issues introduced by reranker-based retrieval and improving retrieval completeness and ranking quality in complex document scenarios. The system has been deployed in production environments at several **top-tier financial institutions**, markedly enhancing operational efficiency.

SELECTED AWARDS

NVIDIA Nemotron Model Reasoning Challenge

June 2026

Silver Medal; ranked 119/4182 globally on the final Kaggle leaderboard.

China National Scholarship

November 2025

Highest academic honor for undergraduates in China (Top 0.2%).

China Undergraduate Mathematical Contest in Modeling

November 2025

First Prize (Team Leader, Top 3%).

Hunan Province Outstanding Student (High School)

April 2023

Awarded to the top 0.1% of high school students in Hunan Province.

TECHNICAL SKILLS & SERVICE

Languages: Chinese (native), English (CET-4: 522, CET-6: 508)

Programming Languages: Python, C/C++, L^AT_EX, Markdown

Tools: Docker, SSH, tmux, Ubuntu

Libraries & Frameworks: ms-swift, vllm, verl, llamaindex, transformers, torch, sklearn, pandas, numpy

Academic Service: Independent Reviewer for *Finance Research Letters*